

Yixin Huang

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EDUCATION

University of California, San Diego, CA

Master of Science in Computer Science

Sep 2025 - Present

University of California, Berkeley, CA

Bachelor of Arts in Computer Science and Applied Mathematics

Aug 2021 – May 2025

GPA: 3.86/4.00

HIGHLIGHTED EXPERIENCES

Hao AI Lab

May 2025 – Present

Research Assistant

- Integrated Doom environment into the Lmgame Bench framework, enabling LLM agent evaluation with harnesses such as memory and perception modules
- Authored a technical blog post demonstrating how to use Lmgame Bench as an evaluation framework for both single-player and multi-player LLM agents
- Built VideoScience-Bench, a scientific video evaluation benchmark assessing physical and chemical correctness of video generation models with VLM-as-a-Judge scoring
- Integrating Sokoban and Tetris game environments into Nvidia's NeMo Gym, enabling large-scale reinforcement learning training and evaluation

Perceptis AI

Jun 2025 – Aug 2025

Software Engineering Intern

- Built an LLM-based pipeline that converted natural language slide descriptions into consulting-grade PowerPoint decks, significantly reducing HITL team effort
- Designed and implemented internal MCP tooling using FastMCP, enabling seamless integration of Claude with tools like custom RAG search over company-specific data
- Conducted exploratory supervised fine-tuning (SFT) on Gemini-2.5 Flash via Vertex AI using Slideworks consulting slides to evaluate potential improvements in model performance for automated slide generation

Emotect AI

Mar 2025 – May 2025

Intern

- Led data curation and schema design for a proprietary emotional classification dataset, incorporating Core Memory Events (CME), Emotion Lexicon (EL), and contextual attributes like valence, intensity, and time decay
- Built a pipeline using LLMs and web scraping to generate and clean thousands of emotional memory samples across diverse cultural and psychological contexts
- Enhanced emotional reasoning of LLaMA model by fine-tuning with Unsloth and LoRA adapters, resulting in improved accuracy and nuanced emotional interpretation across different categories

Sky Computing Lab – UC Berkeley

May 2024 – May 2025

Undergraduate Research Assistant

- Engineered a scalable data generation pipeline for Openfunctions-v3, producing thousands of multi-turn function-calling training data using LLM-based programs with advanced prompt engineering, demonstrating the effectiveness of synthetic data generation for model training
- Spearheaded data cleaning and refinement for BFCL-v3, an open-source leaderboard for evaluating multi-turn and multi-step function-calling abilities of LLMs, accompanied by a widely-read blog post that gained recognition from tech companies including Meta, Salesforce, and Alibaba
- Enhanced LLM evaluation by refining LLM-as-a-Judge to align more closely with human preferences, contributing to the development of Arena-Hard-v2, a rigorous benchmark for assessing LLMs' conversational abilities

SKILLS

Languages: Python, C++, Rust, Java, SQL, Bash

ML & LLM: PyTorch, Transformers, LLMs, LLM Agents, SFT, RL (NeMo RL, Verl), RAG

Systems & Distributed Training: CUDA, NCCL, NVLink, InfiniBand, MPI, DDP, FSDP, Tensor Parallelism, Pipeline Parallelism, Expert Parallelism, Sequence Parallelism

Training & Efficiency: Mixed Precision, Gradient Checkpointing, LoRA/PEFT

Inference & Serving: vLLM, SGLang, KV Cache, Speculative Decoding, Batching

Evaluation: LLM-as-a-Judge, VLM-as-a-Judge

Backend: FastAPI, PostgreSQL, Docker, RabbitMQ

Tools: Linux, Git, Streamlit, Gradio